MILITARY SPECIFICATION

MAGNESIUM OXIDE, CALCINED

(FOR USE IN AMMUNITION)

1. SCOPE

1.1 This specification covers the requirements for one type of calcined magnesium oxide to be used in Aerial Mines.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-D-723 - Drums, Fiber

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

FSC: 1345

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CODE OF FEDERAL REGULATIONS

49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for copies should cite "49 CFR 71-90 (Revised, 1956).")

CONSOLIDATED CLASSIFICATION COMMITTEE

CONSOLIDATED FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

AMERICAN TRUCKING ASSOCIATIONS

NATIONAL MOTOR FREIGHT CLASSIFICATION

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, Washington 6, D.C.).

3. REQUIREMENTS

3.1 The material shall be Magnesium Oxide which conforms to the following requirements:

PROPERTY	Min.	Max.	Applicable Para.
Magnesium, as magnesium oxide (MgO)	97.2%	-	4.3.5
Calcium, as calcium oxide (CaO)	-	1.00%	4.3.4
Silica, as silica dioxide (SiO_2)	-	0.34%	4.3.6
Iron, as iron oxide, (Fe_2O_3)	-	0.18%	4.3.2
Aluminum, as aluminum oxide, (Al_2O_3)	-	0.15%	4.3.9
Loss on Ignition		10.0%	4.3.3
Iodine Number, m.e.q./100 grams	85		4.3.1
Bulk density, loose	8.0 lb/ft^3		4.3.7
Screen Size:			4.3.8
Passes 325 mesh, wet	99.5%		
Passes 200 mesh, wet	99.8%		

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3.2 Workmanship.-The magnesium oxide shall be a fluffy, white powder and shall be free of lumps and foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein.

4.1.1 Contractor quality assurance system.-If the contractor desires to utilize a quality assurance system, which is at variance with the quality assurance provisions of 4.2 and 4.3 and other documents referenced herein, he shall submit a written description of the system to the contracting officer for approval prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection, (Measuring and Testing equipment), and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance provisions required by the provisions of 4.2 and 4.3 and other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedures of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any changes to the written procedure which affect the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

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a. A statement that the lot complies with all requirements and provisions of the approved current written description of the material.

b. Specification number and date, together with an identification and date of changes.

c. Certificates of analysis on all material used directly by the contractor when such material is controlled by Government specification, shall be made available upon request by the contracting officer.

d. Quantity of product in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the cource of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 First article inspection

4.1.3.1 Submission.-The contractor shall submit a first article quantity as designated by the contracting officer for evaluation in accordance with the provisions of 4.1.3.2 (see 6.1). The first article sample shall consist of 500 grams of the material selected randomly in a manner similar to the method described in 4.2.2. The sample submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.1.3.2 Inspections to be performed.-The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.2 and 4.3 of this specification and any or all requirements of the applicable drawings.

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4.1.3.3 Rejection.-If the sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.2 Inspection provisions

4.2.1 Lot formation.-A lot of magnesium oxide shall consist of that quantity produced, by one manufacturer, in not more than 24 consecutive hours of continuous production, with no change in facilities, materials, process or specifications which would change the chemical or physical properties of the composition. In the event the process is a batch operation, each batch shall constitute a lot. The product shall be submitted for inspection in accordance with Standard MIL-STD-105.

4.2.2 Sampling

4.2.2.1 Primary samples.-Ten percent or a minimum of 25, whichever is less, of the containers comprising each lot submitted shall be selected at random. A representative one-pint sample from each selected container shall be taken in the following manner: Insert a closed sampler tube to maximum depth, at an oblique angle, then open and agitate to fill the tube. Each sample shall be placed in a clean, dry, glass jar, which shall be labeled to identify the container, lot number, contract number, and material name.

4.2.2.2 Composite samples.-The primary sample shall be divided into two groups of equal size, and a composite sample shall be prepared from each group. Each composite sample shall be placed in a clean, dry container, sealed, and labeled in a manner that will identify the sample with the contract number, material name, containers, and lot represented. Each composite sample shall be subjected to all chemical and physical tests specified. If either sample fails to meet the requirements specified in 3.1, the lot shall be rejected.

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4.3 Test Methods and Procedures

NOTE: Any deviations from the test procedures specified herein shall require prior approval by the procuring activity.

4.3.1 Iodine number, Major defect, Defect Code No. 01001

4.3.1.1 Reagents: Iodine solution in anhydrous carbon tetrachloride, A.C.S. reagent grade, 0.10 normal (N) solution.
Standard sodium thiosulfate solution, 0.10N.
Potassium iodide solution in 75 percent ethanol, 0.03N.

4.3.1.2 Apparatus: Automatic pipette, 100 ml, gravity fed from iodine solution reservoir equipped with a drying tube on the air inlet.
Burrell "wrist-action" shaker, or equal.
Pipette, 20 ml, stopcock-type.

To maintain adequate accuracy and precision in this method, which is dependent on the measurement of the differences between two relatively large quantities, follow closely the prescribed details of technique, which are designed to minimize the introduction of moisture from the atmosphere and from absorbed films on glassware, and to avoid uncompensated absorption of iodine on glass surfaces. Glassware must be cleaned with acid to remove minute deposits of basic magnesium compounds.

The determination shall be made as follows: Weigh rapidly a 2.0 gm sample and transfer the sample quantitatively, through a powder funnel, into a 100 ml cylinder. Rinse the automatic pipette with 0.10N iodine solution and discard the rinse. Immediately refill the pipette, and add 100 ml of the iodine solution to the sample. Prepare a blank to be run in parallel with the sample. The rinse need not be repeated if the solution is added to subsequent samples without delay. Secure the glass stopper in each cylinder with a clip, and center the cylinder horizontally in the mounting clamp on the shaker. Shake the cylinder at maximum rate for 30 minutes, remove from the shaker, and allow the contents to settle for 10 minutes. From the top third of the clear supernatant liquid in each cylinder, transfer a 20 ml aliquot, using a clean, dry stopcock pipette (the pipette may be air dried with a rubber bulb between determinations) to a 250 ml Erlenmeyer flask, or equal, containing 50 ml of 0.03N alcoholic

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potassium iodide solution, and titrate with standard 0.10N sodium thiosulfate.

4.3.1.3 Calculation:

Calculate the iodine number (mg-equivalent per 100 gm) as follows:

$$\text{Iodine number} = \frac{5\ (A-B)\ N\ (100)}{W}$$

Where:

A = ml sodium thiosulfate solution used for blank.
B = ml sodium thiosulfate solution used for sample.
N = normality of sodium thiosulfate solution.
W = weight of sample, gm.

4.3.2 Iron, as iron oxide

4.3.2.1 Iron oxide, Major defect, Defect Code No. 02001

4.3.2.1.1 Reagents: H_2SO_4 solution, Reagent grade, 50 percent.
Hydroxylamine hydrochloride, 10 percent, freshly made.
O-Phenanthroline solution, 0.25 percent.
Sodium citrate, Reagent grade, ($Na_3C_6H_5O_7.2H_2O$) 25 percent solution.
HCl, Reagent grade, concentrated
Nitric Acid (HNO_3), Reagent grade, concentrated.
Bright iron wire, primary standard grade.

4.3.2.1.2 Apparatus: Beckman spectrophotometer, Model DU, or equal, with 3 matched 1cm corex cells, set up for use with tungsten lamp and red sensitive phototube, without filter.

Determination of the absorptivity shall be made as follows: Weigh accurately 0.05 gm of bright iron wire (an equivalent weight of any other recognized primary standard, with appropriately modified preparation procedure, may be substituted for the iron wire standard) and place it in a 1 liter volumetric flask, add 2 ml of concentrated HCl and a few ml of hot distilled water. When all the iron is dissolved, fill the flask to volume, and mix.

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Transfer 1.00, 3.00, 5.00, 7.00 and 9.00 ml portions of the standard iron solution to a series of 100 ml volumetric flasks, and prepare a blank. Dilute each to approximately 70 ml and determine the absorbency (A_s) by the procedure described below for analysis of the sample. The initial calibration should consist of four complete determinations, two on each of two days, preferably one by each of two analysts. The calibration should be checked by two additional single determinations for each new bottle of solid O-phenanthroline reagent. Calculate the concentractions, plot them against the measured absorbances, and draw the line of best fit. A line that deviates significantly from the zero intercept or shows curvature indicates error. Calculate numerically the absorptivity as follows:

$$a_s = \frac{\Sigma(A_s bC) - (\Sigma A_s)\left(\frac{\Sigma bC}{n}\right)}{\Sigma(bC)^2 - \frac{(\Sigma bC)^2}{n}}$$

Where:

C = concentration of iron in diluted aliquot, gm per ml = $\frac{WV}{100{,}000}$.
W = weight of primary standard, gm.
V = volume of standard solution used, ml.
A_s = absorbance.
b = length of light path through the cell, cm.
n = number of determinations.
a_s = absorptivity, ml per gm-cm (2.03×10^5 ml per gm-cm for Beckman spectrophotometer, model DU.).

The percent of iron oxide determination shall be made as follows: Weigh 1 gm of sample and transfer it to a 50 ml beaker. Add 20 ml of distilled water, dissolve the sample by dropwise addition of concentrated HCl until the solution is just acid to litmus, then add 1 ml of acid in excess, and prepare a blank to run parallel to the sample. Bring to a gentle boil to insure complete solution, then cool. Transfer the solution to a 100 ml volumetric flask, dilute to mark, and mix. Pipette an aliquot (usually 5 ml) of this solution to a second 100 ml volumetric flask, and dilute to approximately 70 ml with distilled water. Treat both sample and blank as follows: Add 1 ml of freshly prepared hydroxylamine hydrochloride solution, mix well, and let the mixture stand for at least 15 minutes. Add 1 ml of freshly prepared hydroxylamine hydrochloride solution, mix well, and let the mixture stand for at least 15 minutes.

Add 10 ml of 0.25 percent O-phenanthroline solution and mix well. Add 1 ml of 25 percent sodium citrate solution, dilute to volume,

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mix well, and allow to stand for at least 30 minutes at not less than 20 degrees C. Using the blank for comparison, determine the absorbance (A_S) of the sample solution with spectrophotometer at a wavelength of 510 millimicrons and a slit width of 0.18 mm.

4.3.2.1.3 Calculation:

Calculate the percent of Fe_2O_3 as follows:

$$Fe_2O_3, \text{ percent} = \frac{79.8\ (10^6)\ A_S}{55.8 a_S b w v} = \frac{14.3\ (10^5)\ A_S}{a_S b w v}$$

Where:

A_S = absorbance of sample solution.
a_S = absorptivity, ml per gm-cm.
b = cell length, cm.
v = volume of aliquot, ml.
w = sample weight, gm.

4.3.2.2 Iron oxide - Alternate method, Major defect, Defect Code No. 02002

4.3.2.2.1 Apparatus and Reagents.-All catalog numbers refer to Fisher Scientific Catalog 59, unless otherwise specified. Equivalent materials can be obtained from alternate suppliers. Reagent solutions can be purchased or prepared as described.

1 Beckman Spectrophotometer, Model DU, 14-383
1 pH meter, Beckman Model N-1 11-505-301
Ferron reagent, 0.2% solution. Dissolve 0.40 g reagent grade ferron (8-hydroxy-7-iodo-5-quinolinesulfonic acid) in deionized water and dilute to 200 ml (complete dissolution is obtained by heating to 80°C). 4449
Ammonium acetate buffer, 10% solution. Dissolve 50.0 g of reagent grade ammonium acetate in deionzied water and dilute to 500 ml. A-637
Nitric acid, 1:9 solution. Add 10 ml concentrated nitric acid to 90 ml deionized water. A-200
Magnesia solution, 1 g MgO/100 ml concentrated nitric acid and dilute to 1 liter with deionized water. M-51
Stock iron standard, 20µg Fe_2O_3/ml. Dissolve 0.0140 g of electrolytic iron wire in 10 ml concentrated hydrochloric acid and 5 ml concentrated nitric acid and evaporate almost to dryness. Dissolve in deionized water and dilute to 1 liter. 15-550
Stock aluminum standard, 20 µg Al_2O_3/ml. Dissolve 0.0106 g reagent grade aluminum turnings in 10 ml concentrated hydrochloric acid and evaporate almost to dryness.

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Dissolve in deionized water, dilute to 1 liter in a volumetric flask and mix thoroughly. Hot plate, gas two burners. 4-220
Buffer solution pH 4.00 11-505-225
Buffer solution, pH 10.00 11-505-234
Pipets, volumetric, 10, 20, 25, 30 ml. 13-649
Buret, Schellbach automatic zero, 3-750

4.3.2.2.2 Preparation of standards:

1. Into four 150-ml Griffin beakers introduce, by pipet, 5, 15, 25, and 50 ml of the aluminum stock solution (100, 300, 500 and 1000 µg Al_2O_3).

Into four 150-ml Griffin beakers introduce 5, 15, 25 and 50 ml of the iron stock solution (100, 300, 500 and 1000 µg Al_2O_3).

Use an empty 150-ml beaker for a reagent blank. This can be used for both standard series. Mark each beaker distinctly to identify its contents.

2. To each of the beakers, add the following reagents in the order and quantities given:

25 ml Magnesia solution
4 ml 1:9 HNO_3
Sufficient 10% ammonium acetate solution to bring the pH to 5.2 (This will require about 25 ml) using a pH meter.

3. Transfer the contents of each beaker to a 200 ml volumetric flask, rinsing the beaker thoroughly and adding the rinse water to the volumetric flask. Add 10 ml of ferron reagent, dilute to the mark with deionized water and mix thoroughly. Allow the solutions to stand for 15 minutes for color development.

4. Place each of the standards and the blank in a 1.00 cm absorption cell, and using the blank as reference, measure the absorbance of each iron solution at 370 mµ and 600 mµ ; and each aluminum solution at 370 mµ, with the Bechman DU spectrophotometer as described in 4.3.2.2.3 and prepare standard curves.

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4.3.2.2.3 Procedure:

The determination shall be made as follows:

a. Weigh a 1.000 g sample and transfer to a 250 ml beaker, cover with 5 ml of deionized water, and add 6-10 ml of concentrated hydrochloric acid. Warm on a hot plate at low heat until the sample has dissolved. Transfer the solution to a 100 ml volumetric flask, rinsing the beaker thoroughly. Dilute to the mark.

b. Transfer a 25 ml aliquot of the above solution by pipet to a 150 ml beaker and add 4 ml of 1:9 nitric acid to oxidize all iron. To a second 150 ml beaker, add 25 ml of standard MgO solution and carry through all subsequent steps as a reagent blank.

c. Place the electrodes of a pH meter into the solution and, by means of a 50 ml buret, add 10% ammonium acetate buffer until a pH of exactly 5.2 is obtained. Rinse the electrodes into the beaker with deionized water. Rinse the solution into a 200 ml volumetric flask with deionized water.

d. Add 10 ml of ferron reagent by pipet and dilute to the mark with deionized water. Invert the volumetric flask repeatedly to mix. Allow to stand for 15 minutes for color development.

e. Fill a 1.000 cm Beckman cell with the sample solution, and a second with the blank. Place the cells in the carrier of the Beckman DU spectrophotometer.

f. Put the tungsten lamp in place, and make suitable settings for measuring at 370 mμ as described below.

Instrument Settings	Type of Instrument			
	With Photomultiplier		Without Photomultiplier	
Wavelength, mμ	370	600	370	600
Slit width, mm	0.2	0.025	0.2	0.05
Photo tube	P-M	P-M	Blue	Blue
Filter	Corex	None	Corex	None
Resistor, megohms	22	22	2000	2000
Check switch	0.1	0.1	0.1	0.1
P-M sensitivity	3	3	-	-

g. Turn the tungsten lamp on and the selector switch to "check". With the shutter switch closed, zero the galvanometer with the dark current control.

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h. Place the blank solution in the light path, open the shutter switch and again zero the galvanometer, this time using the sensitivity control.

i. Turn the selector switch to 0.1 and place the sample solution in the light path. Balance the galvanometer with the transmittance control. Read the absorbance from the scale, and record as "absorbance at 370 mμ ".

j. Make the necessary settings for measuring at 600 mμ. Repeat steps g through i, and record this reading as "absorbance at 600 mμ".

k. From the standard curve for aluminum at 370, using the "absorbance at 370 mμ", obtain and record "total uncorrected aluminum".

l. Obtain the iron concentration from the iron standard curve at 600 using the "absorbance at 600 mμ".

m. Using the iron concentraction obtained in step l obtain the "iron absorbance at 370mμ" from the iron standard curve at 370 mμ.

n. Using the iron absorbance at 370mμ, obtain from the aluminum standard curve, the equivalent aluminum concentraction. Subtract this concentration from the "total uncorrected aluminum", (step k). Record as "true aluminum".

4.3.2.2.4 Calculations:

Since both the standard and the sample are diluted to the same volume, the results are directly comparable. Thus, the amount of Fe_2O_3 and of Al_2O_3 are given by the following equations:

$$\frac{\mu g\ Fe_2O_3\ found}{g\ sample} \times \frac{1\ g}{1{,}000{,}000\ \mu g} \times \frac{100\ ml\ dilution}{25\ ml\ aliquot} \times 100 = \%Fe_2O_3$$

$$\frac{\mu g\ Al_2O_3\ (true\ Al)}{g\ sample} \times \frac{1\ g}{1{,}000{,}000\ \mu g} \times \frac{100\ ml\ dilution}{25\ ml\ aliquot} \times 100 = \%Al_2O_3$$

When taking a 1.000 g sample, diluting to 100 ml and taking a 25 ml aliquot the claculations become

$$\mu g\ Fe_2O_3\ found \times 0.0004 = \%\ Fe_2O_3$$

$$\mu g\ Al_2O_3\ found\ (true\ Al) \times 0.0004 = \%\ Al_2O_3$$

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4.3.3 Loss on Ignition, Major Defect, Defect Code No. 03001

4.3.3.1 Apparatus

1 Muffle furnace (1600°F)
1 Analytical balance
2 Porcelain crucibles, Coors No. 1 (tall form)
1 Desiccator

4.3.3.2 Procedure:

The determination shall be made as follows:

Ignite a crucible for one hour in the muffle furnace at 1600°F. Remove the crucible and allow to cool to room temperature in a desiccator. Weight the crucible and record the weight to the nearest 0.0001 g. Place approximately one (1) gram of sample in the crucible and weigh to the nearest 0.0001 g. Ignite in the muffle at 1600°F for one hour. Remove and cool to room temperature in a desiccator. Weigh and record the weight.

4.3.3.3 Calculation:

Calculate the ignition loss from the following formula:

$$\frac{\text{weight loss (g)}}{\text{sample weight (g)}} \times 100 = \% \text{ ignition loss}$$

NOTE: Prepare an additional amount of ignited sample for use in 4.3.5.

4.3.4 Calcium, as calcium oxide

4.3.4.1 Calcium oxide, Major defect, Defect Code 04001

4.3.4.1.1 Reagents: Sulfuric acid, 25 percent reagent grade.
Ethanol, Reagent grade.

4.3.4.1.2 Procedure:

The determination shall be made as follows: Except for 0.5 gm (to be used in 4.3.5), transfer the remainder of the ignited sample obtained in 4.3.3 to a tared 30 ml beaker and weigh. Add

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4 ml of water, then stir and add HCl dropwise until the sample is just dissolved. Dilute to 10 ml with water and filter through Whatman #40 filter paper, or equal, into a 250 ml beaker. Add to the filtrate 100 ml of ethanol and 40 ml of 25 percent H_2SO_4. Stir well and allow to stand over night. Filter the liquid through an ignited (400 to 500 degrees C.) and tared Selas crucible, or equal. Transfer and wash the precipitate with a mixture of 2 volumes of ethanol to 1 volume of 25 percent H_2SO_4. Dry the crucible and precipitate at 105 degrees C for 30 minutes with the crucible supported on a ceramic plate (to prevent reaction of the retained acid with the metal shelf of the oven), then ignite at 400 to 500 degrees C for thirty minutes. Cool in a desiccator and weigh.

4.3.4.1.3 Calculation:

Calculate percent of CaO as follows:

$$\text{CaO, percent} = \frac{56.08A}{136.14W} = \frac{.412A}{W}$$

Where:

A = weight of calcium sulfate, gm.
W = weight of ignited sample, gm.

4.3.4.2 Calcium oxide (alternate method), Major defect, Defect Code No. 04002

4.3.4.2.1 Apparatus and Reagents

Analytical balance
Crucibles, Gooch, porcelain, No. 3
Asbestos fiber, analytical grade
Hydrochloric acid, concentrated, reagent grade
Ammonium hydroxide, concentrated, reagent grade
6N Sulfuric acid (1:6)
10% Sulfuric acid solution
0.1 N Potassium permanganate, standardized
4% Ammonium oxalate solution
0.1% Ammonium oxalate solution
5% HCl-NH_4Cl solution--50 g ammonium chloride plus 50 ml conc. hydrochloric acid diluted to 1 liter with deionized water.
Methanol, absolute, reagent grade
Methyl red indicator

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4.3.4.2.2 Procedure:

The determination shall be made as follows: Weigh approximately 3 g of sample on the analytical balance to the nearest milligram and transfer to a 250 ml beaker. Add 10 ml deionized water and 20 ml concentrated HCl, with swirling, and evaporate the sample to dryness. Cool the beaker and then add 5 ml 6N sulfuric acid. Cover the beaker with a watch glass and heat until all salts are dissolved. Cool slightly and add 50 ml absolute methanol with thorough mixing. Filter the solution through a prepared Gooch crucible with an asbestos mat, washing the beaker with five 10.0 ml portions of methanol. Transfer the asbestos mat with the precipitate to the original beaker, washing with 35 ml of 5% HCl-NH_4Cl solution. Continue washing the crucible with deionized water until the volume in the beaker is about 100 ml. Add 3 drops of methyl red indicator and heat to boiling. While still hot add 25 ml 4% ammonium oxalate solution with stirring and then add concentrated ammonium hydroxide dropwise until the solution turns yellow. Allow to stand in a warm place for one-half hour. Filter through another prepared Gooch crucible with an asbestos mat, washing the beaker with five 10.0 ml portions of 0.1% ammonium oxalate solution. Transfer entire crucible with the precipitate to a 250 ml beaker, add 150 ml 10% sulfuric acid, and heat almost to boiling. Titrate while hot with 0.1 N potassium permanganate to a permanent pink endpoint.

4.3.4.2.3 Calculation:

Calculate the calcium oxide concentration as follows:

$$\% \text{ CaO} = \frac{\text{ml } KMnO_4 \times \text{N of } KMnO_4 \times 0.02804 \times 100}{\text{sample weight}}$$

4.3.5 Magnesium oxide, Major defect, Defect Code No. 05001

4.3.5.1 Reagents: Standard HCl solution, 0.0N.
Standard NaOH solution, 1.0N.
Methyl orange indicator

The determination shall be made as follows: Weigh accurately the appropriate 0.5 gm of ignited sample, retained in 4.3.4, into a 250 ml Erlenmeyer flask, or equal. Dissolve the sample in 50.0 ml of standard 1.0N HCl. Back titrate the excess acid with standard 1.0N NaOH, using methyl orange indicator.

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4.3.5.2 Calculation:

Calculate the percent of MgO as follows:

$$\text{MgO, percent} = \frac{(An_a - BN_b)\ 20.16}{1000W}(100) - 40.32\left(\frac{C}{56.08} + \frac{D}{53.0}\right)$$

$$= 2.016\ \frac{(AN_a - BN_b)}{W} - 0.72C - 0.76D$$

Where:

A = volume of standard HCl, ml.
B = volume of standard NaOH, ml.
C = CaO, percent (from 4.3.4)
D = Na_2CO_3, percent (see note below)
Na = normality of standard HCl.
Nb = normality of standard NaOH.
W = weight of ignited sample.

NOTE: The determination of Na_2CO_3 shall be made as follows: Weigh 5.0 gm of sample, add 75 ml of water, and boil for 5 minutes. Filter with mild vacuum through an 80 mm funnel (Buchner or equivalent), fitted with Whatman No. 2 filter paper. Compare the filtrate with distilled water, and if any cloudiness is noted, filter again through a Selas No. 3001 crucible or equivalent. Cool the filtrate, transfer to a 100 ml volumetric flask, dilute to mark, mix, and titrate a 20 ml aliquot with 0.10N HCl using methyl red as indicator. Calculate the percent Na_2CO_3 as follows:

$$Na_2CO_3\text{, percent} = \frac{5\ AN\ (53.0)(100)}{\text{wt sample}\ (1000)} = \frac{AN\ (26.5)}{\text{wt sample}}$$

Where:

A = volume of standard HCl ml.
N = normality of standard HCl.

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4.3.6 Silica (SiO_2), Major defect, Defect Code No. 06001

4.3.6.1 Apparatus and Reagents

1 Muffle furnace (1600°F)
1 Analytical balance
1 Hot plate
1 Laboratory oven (110°C)
1 Desiccator
Porcelain crucibles, Coors No. 1 (tall form)
Concentrated hydrochloric acid, reagent grade
Acetic anhydride, reagent grade
Filter paper, ashless

4.3.6.2 Procedure:

The determination shall be made as follows: Weigh approximately one (1) gram of the sample to the nearest 0.0001 gram. Transfer the material to a 250 ml beaker and add 5 ml deionized water and 20 ml concentrated hydrochloric acid. Place beaker on a hot plate, break up all gritty particles with a stirring rod, and cover with a watch glass. Evaporate the sample to dryness and add 10 ml acetic anhydride and 10 ml deionized water to the slightly cooled residue. Heat on the hot plate. A mild reaction indicates sufficient heating. At this point place the sample in a warm place until the reaction ceases. Add 4 ml of 1:1 hydrochloric acid and 20 ml of deionized water. Mix thoroughly. Filter the sample through Whatman No. 30 ashless filter paper. Wash the residue with 25 ml of hot acidified water (5 ml concentrated hydrochloric acid to 1 liter deionized water) and then wash thoroughly with hot deionized water. Transfer the filter paper to a previously ignited, tared crucible. Char the filter paper with a blast lamp and ignite for one hour in the muffle furnace at 1600°F. Cool and reweigh the crucible. The weight difference is the weight of the precipitate.

4.3.6.3 Calculation:

Calculate the silica concentration as follows:

$$\frac{\text{wt ppt - wt filter ash*}}{\text{wt sample}} \times 100 = \% \ SiO_2$$

*Printed on filter paper box

4.3.7 Bulk density, loose, Major defect, Defect Code No. 07001

4.3.7.1 Apparatus

1 Laboratory screen, 20 mesh

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1 Cylinder of known volume (100 ml)
1 Powder funnel, 6 inch diameter
1 Triple-beam balance

4.3.7.2 Procedure:

The determination shall be made as follows: Weigh a 100 ml cylinder (a 250 ml cylinder which has been cut off at the 100 ml mark) to the nearest 0.01 g. Position the powder funnel above the cylinder so that its bottom tip is one inch above the cylinder rim. Screen the sample through a 20 mesh sieve to avoid any lumps. Slowly shake the powder from a spatula so that it falls onto the side walls of the funnel. Avoid any vibration. Fill the cylinder to heaping. Carefully scrape the cylinder rim with the spatula so as to level the contents in a single stroke. Carefully remove any material clinging to the outside of the cylinder. Reweigh the filled cylinder and calculate the bulk density with the following formula:

4.3.7.3 Calculation:

$$\frac{\text{(Wt. of full cylinder - Wt. of empty cylinder)}}{\text{cylinder volume}} \times 62.4 = \text{Bulk Density (lb/cu ft)}$$

4.3.8 Screen Size, Major defect, Defect Code No. 08001

4.3.8.1 Apparatus and Reagents

1 U.S. Standard Screen No. 325 with pan
1 U.S. Standard Screen No. 200 with pan
1 Triple-beam balance
1 Analytical balance
1 Paint brush, 1/2 inch wide by 1 5/8 inch long, with Pure bristles
1 Laboratory oven
1 Graduate, 500 ml
1 Separator funnel, 1 l (globe-shaped)
Tubing, Tygon
Acetone

4.3.8.2 Wet Screen Test

Weigh 10 $\pm$ 0.1 grams of sample on a triple-beam balance and transfer to the 325 mesh screen (and 200 mesh screen). Clamp the screen under the globe-shaped funnel which has a tygon tube (1/2 foot to 1 foot in length and 1/4 inch in inside diameter) attached. Measure 300 ml of water into the funnel. Open the funnel and allow the water to flow on the magnesia and screen while the magnesia is being brushed with the 1/2 inch brush.

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Regulate the flow of water so that the step above will take 2.0 ± 0.1 minutes and so that as much of the sample as possible is brushed through the screen. Wash the magnesia from the brush on to the screen with the last 40 ml of water. Rinse the screen and residue with acetone and dry in an oven at 110°C. Carefully brush out the residue and weigh on an analytical balance.

4.3.8.3 Calculation:

$$\frac{\text{wt of residue X 100}}{\text{sample, wt}} = \% + \text{(screen size)}$$

$$100.00 - \% + \text{screen} = \% - \text{(screen size)}$$

NOTE: The particle size of the material is determined by screening a weighed representative sample through an appropriate sieve (U.S. Standard Screen Series). The material passing through or the residue remaining on a screen is reweighed to give the fraction passing through or held by the screen (reported as %- or %+ screen size respectively). Fine powders are "washed" through using a brush and water.)

4.3.9 Aluminum, as aluminum oxide, Major defect, Defect Code No. 09001

4.3.9.1 Apparatus and Reagents

1 Analytical balance
1 Hot plate
1 Muffle furnace (1600°F)
1 Laboratory oven (110°C)
1 Desiccator
Hydrochloric acid, reagent grade
Ammonium Hydroxide, reagent grade
Methyl Red indicator solution
Filter paper, Whatman No. 30
2 Porcelain crucibles, Coors No. 1, tall form
Ammonium nitrate, reagent grade, 1% solution

4.3.9.2 Procedure:

The determination shall be made as follows: Weigh approximately two (2) grams of material to the nearest 0.0001 g. Transfer the material to a 250 ml beaker. Add 5 ml distilled water and 20 ml concentrated hydrochloric acid with caution. Digest on a hot plate until solution is complete. Break up any lumps with a stirring rod and cover the beaker with a watch glass. Evaporate the solution to dryness.

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Cool the sample in air and add 5 ml concentrated hydrochloric acid. Warm for a few minutes on a hot plate, then add 50 ml of hot acidified water (5 ml of 1:1 HCl in 1 liter of distilled water). Filter the solution through No. 30 paper, into a 400 ml beaker, washing the residue with hot water until the filtrate reaches a volume of about 250 ml. To the filtrate add 5 g ammonium chloride, 3 drops of methyl red indicator and heat to boiling. Add concentrated ammonium hydroxide slowly, finally drop by drop until just alkaline (yellow to methyl red). Add a pinch of ashless filter pulp and boil for 3 minutes. Let stand for 15 minutes to coagulate the hydrous oxides. Filter the solution through No. 30 paper. Wash the residue 4-5 times with hot 1% ammonium nitrate solution. Transfer the filter paper to a previously ignited and tared crucible, dry in an oven (100°C) for one hour. Char the filter paper with a low heat and finally ignite at red heat until no carbon remains. Ignite the crucible in the muffle for one hour. Cool in a desiccator and reweigh the crucible. Difference in weight is the precipitate.

4.3.9.3 Calculation:

Calculate the iron oxide and aluminum oxide concentration from the following formula:

$$R_2O_3 = \% \ (Fe_2O_3 + Al_2O_3)$$

$$\%R_2O_3 = \frac{(\text{wt of ppt} - \text{ash of paper}^*)}{\text{sample wt}} \times 100$$

Using the iron concentration as determined in 4.3.2, calculate the Al_2O_3 concentration as follows:

$$\% \ Al_2O_3 = \% \ R_2O_3 - \% \ Fe_2O_3$$

* Ash of each filter paper is printed on box of filter papers.

5. PREPARATION FOR DELIVERY

5.1 Packing (see 6.5)

5.1.1 Drums.-Unless otherwise specified in the contract or purchase order, the magnesium oxide shall be packed in non-returnable fiber drums in accordance with Specification PPP-D-723, (Type III, Grade A). The drums shall be finished with a normal 0.004 inch thick polyethylene liner, properly heat sealed or otherwise closed to afford protection to the contents equivalent to that given by a heat seal closure.

5.1.2 Level C.-The magnesium oxide shall be packed to afford protection against damage during direct shipment from supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules and Container Specifications for rail shipments or with National Motor Freight

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Rules and Container Specifications for truck shipments as applicable.

5.2 Marking.-In addition to any special markings required by the contract or purchase order, containers shall be marked in accordance with MIL-STD-129 and Interstate Commerce Commission requirements specificed in Code of Federal Regulations 49 CFR 71-90. Marking shall include, but not limited to, the following information:

a. Manufacturers name
b. Product designation
c. Lot number
d. Date of manufacture
e. Number of this specification

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, class and date of this document, and provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.3 Intended use.-The magnesium oxide covered in this specification is intended for use in Aerial Mines.

6.4 One type of magnesium oxide which has been found satisfactory is Michigan 1782 Magnesia made by Michigan Chemical Corp., Chicago, Illinois.

6.5 The packing described in 5.1.1 is advisory rather than mandatory. If the supplier wishes to deviate from the provisions in 5.1 he shall obtain the approval of the procuring activity.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1345-A-250

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